**FY-927**

Reg. No. : .....

Name : .....

# FIRST YEAR HIGHER SECONDARY
# IMPROVEMENT/SUPPLEMENTARY EXAMINATION, OCTOBER 2026

Part – III

Time : 2 Hours

## MATHEMATICS (SCIENCE)

Cool-off time : 15 Minutes

Maximum : 60 Scores

***General Instructions to Candidates :***

- There is a 'Cool-off time' of 15 minutes in addition to the writing time.
- Use the 'Cool-off time' to get familiar with questions and to plan your answers.
- Read questions carefully before answering.
- Read the instructions carefully.
- Calculations, figures and graphs should be shown in the answer sheet itself.
- Malayalam version of the questions is also provided.
- Give equations wherever necessary.
- Electronic devices except non-programmable calculators are not allowed in the Examination Hall.

*വിദ്യാർത്ഥികൾക്കുള്ള പൊതുനിർദ്ദേശങ്ങൾ :*

- നിർദ്ദിഷ്ട സമയത്തിന് പുറമെ 15 മിനിറ്റ് 'കൂൾ ഓഫ് ടൈം' ഉണ്ടായിരിക്കും.
- ചോദ്യങ്ങൾ പരിചയപ്പെടാനും ഉത്തരങ്ങൾ ആസൂത്രണം ചെയ്യാനും 'കൂൾ ഓഫ് ടൈം' ഉപയോഗിക്കുക.
- ഉത്തരങ്ങൾ എഴുതുന്നതിനു മുമ്പ് ചോദ്യങ്ങൾ ശ്രദ്ധാപൂർവം വായിക്കണം.
- നിർദ്ദേശങ്ങൾ മുഴുവനും ശ്രദ്ധാപൂർവം വായിക്കണം.
- കണക്ക് കൂട്ടലുകൾ, ചിത്രങ്ങൾ, ഗ്രാഫുകൾ, എന്നിവ ഉത്തരപേപ്പറിൽ തന്നെ ഉണ്ടായിരിക്കണം.
- ചോദ്യങ്ങൾ മലയാളത്തിലും നല്കിയിട്ടുണ്ട്.
- ആവശ്യമുള്ള സ്ഥലത്ത് സമവാക്യങ്ങൾ കൊടുക്കണം.
- പ്രോഗ്രാമുകൾ ചെയ്യാനാകാത്ത കാൽക്കുലേറ്ററുകൾ ഒഴികെയുള്ള ഒരു ഇലക്ട്രോണിക് ഉപകരണവും പരീക്ഷാഹാളിൽ ഉപയോഗിക്കുവാൻ പാടില്ല.

**Answer any 6 questions from 1 to 8. Each carries 3 scores.** **(6 × 3 = 18)**

1. (a) If A and B are two set such that $A \cup B = A \cap B$, then **(1)**

   (A) $A = \phi$ (B) $B = \phi$

   (C) $A = B$ (D) None of these

   (b) For any sets P and Q, prove that $(P \cap Q) \cup (P - Q) = P$. **(2)**

2. (a) Find the value of tan 135°. **(1)**

   (b) A circle of arc length $15\pi$ centimetre makes an angle $3\frac{\pi}{4}$ radian at the centre. Find the radius of the circle. **(2)**

3. (a) Solve the inequality, $\frac{x}{3} > \frac{x}{2} + 1; x \in \mathbb{R}$. **(2)**

   (b) Mark the solution obtained in part (a) to the real line $\mathbb{R}$. **(1)**

4. (a) Write the difference between the LCM and the HCF of 4! 5! 6!. **(1)**

   (A) 24 (B) 120

   (C) 696 (D) 800

   (b) If $^{12}C_r = 495$ and $^{12}P_r = 11880$, then find the value of r. **(2)**

5. (a) Perpendicular distance from the point (6, 7, 8) to the XY-plane is **(1)**

   (A) 6 (B) 7

   (C) 8 (D) 9

   (b) Find the equation of the set of points which are equidistant from the points (1, 2, 3) and (3, 2, –1). **(2)**

6. Find the centre and radius of the circle $x^2 + y^2 - 4x - 8y - 45 = 0$. **(3)**

7. (a) Evaluate $\lim_{x \to 0} \frac{\sin 5x}{x}$ **(1)**

   (b) Evaluate $\lim_{x \to 3} \frac{x-3}{x^2 - 5x + 6}$ **(2)**

8. (a) Find the slope of the line $x - y = 8$. **(1)**

   (b) Two lines $x - 2y = 5$ and $y - 2x = 5$ makes an angle $\theta$ between them. Find $\theta$. **(2)**

**Answer any 6 questions from 9 to 16. Each carries 4 scores.** **(6 × 4 = 24)**

9. (a) If $f(x) = 2x^2 - 5$ and $g(x) = 2x$, then find $(fog)(0)$. **(1)**

   (b) Find the range of $f(x) = x^2 + 2; x \in \mathbb{R}$. **(3)**

10. (a) How many relations can be defined from the set $A = \{x, y, z\}$ to set $B = \{1, 2\}$ ? **(1)**

   (b) Consider the real valued function $f(x) = \begin{cases} 1 - x, & x < 0 \\ 1, & x = 0 \\ x + 1, & x > 0 \end{cases}$

   (i) Draw the graph of $f(x)$. **(1)**

   (ii) Write the domain and range of $f(x)$. **(2)**

11. (a) Find the value of $i^{102}$ and $\frac{1}{i^{25}}$. **(2)**

   (b) Write the modulus of the complex number $\frac{1+i}{1-i}$. **(2)**

12. How many ways can the letters of the word **PERMUTATIONS** be arranged so that

   (a) words start with **P** and end with **S** **(2)**

   (b) vowels come together **(2)**

13. (a) Find the equation of a line passing through the point (1, 4) and (4, 10). **(2)**

   (b) Write its intercepts on axes. **(1)**

   (c) Find the distance of this line from the origin. **(1)**

14. An ellipse passing through (6, 5) having foci (± 6, 0). Find its

   (a) length of major and minor axis **(2)**

   (b) equation **(1)**

   (c) eccentricity **(1)**

15. (a) Write the number of terms in the expansion of $(2x - y)^{99}$. **(1)**

   (b) Using Binomial theorem, show that $6^n - 5n$ always leaves remainder 1 when divided by 25. **(3)**

16. (a) If A is an event of a random experiment with $P(A) = \frac{3}{5}$, then find $P(\overline{A})$. (1)

(b) If a fair coin with 1 marked on one face and 6 on the other face and a fair die are tossed together, then find the probability that the sum of the numbers turn up is 12. (3)

**Answer any 3 questions from 17 to 20. Each carries 6 scores.** **(3 × 6 = 18)**

17. (a) Evaluate $\sec\left(\frac{13\pi}{6}\right)$. (1)

(b) Prove that $\frac{\tan\left(\frac{\pi}{4}+x\right)}{\tan\left(\frac{\pi}{4}-x\right)} = \left(\frac{1+\tan x}{1-\tan x}\right)^2$. (2)

(c) Prove that $(\cos x - \cos y)^2 + (\sin x - \sin y)^2 = 4\sin^2\left(\frac{x-y}{2}\right)$. (3)

18. (a) Write the first three terms of the sequence formed by $a_n = \frac{n}{n+1}$; $n \in \mathbb{N}$, natural number. (2)

(b) The fourth term of a geometric progression is square of its second term and the first term is –3. Find its seventh term. (2)

(c) Find the sum of first 20 terms of the geometric progression $\sqrt{7}, \sqrt{21}, 3\sqrt{7}, \ldots$ (2)

19. (a) Find the derivative of $\cos x$ using first principle. (2)

(b) Find the derivative of

(i) $\frac{x^2}{x+1}$ (2)

(ii) $\sin x \cos x$ (2)

20. (a) Find Mean Deviation of the following data :

3, 9, 5, 3, 12, 10 (2)

(b) Find the Standard Deviation of the following observations :

| $x_i$ | 6 | 10 | 14 | 18 | 24 | 28 | 30 |
|---|---|---|---|---|---|---|---|
| $f_i$ | 2 | 4 | 7 | 12 | 8 | 4 | 3 |

(4)